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Studierichting: Informatica
Oefeningenreeks 4A nr. 41.13

$$\begin{aligned}& \int \frac{x-1}{\sqrt{x^2+5x+6}} dx \\&= \int \frac{\frac{1}{2}(2x+5) - \frac{7}{2}}{\sqrt{x^2+5x+6}} dx \\&= \frac{1}{2} \int \frac{2x+5}{\sqrt{x^2+5x+6}} dx - \frac{7}{2} \int \frac{1}{\sqrt{x^2+5x+6}} dx \\& t = x^2 + 5x + 6 \Rightarrow dt = (2x+5)dx \\&= \frac{1}{2} \int \frac{dt}{\sqrt{t}} - \frac{7}{2} \int \frac{dx}{\left(x + \frac{5}{2}\right)^2 - \frac{1}{4}} \\&= \frac{1}{2} \int t^{-\frac{1}{2}} dt - \frac{7}{2} \int \frac{dx}{\left(x + \frac{5}{2}\right)^2 - \frac{1}{4}} \\&= \frac{1}{2} \cdot \frac{2}{1} t^{\frac{1}{2}} - \frac{7}{2} \ln \left| \left(x + \frac{5}{2}\right) + \sqrt{\left(x + \frac{5}{2}\right)^2 - \frac{1}{4}} \right| + C \\&= \sqrt{x^2+5x+6} - \frac{7}{2} \ln \left| x + \frac{5}{2} + \sqrt{x^2+5x+6} \right| + C\end{aligned}$$

References